

### Exercise 4.1

You are given 12 balls, all equal in weight except for one that is either heavier or lighter. You are also given a two-pan balance to use. In each use of the balance you may put any number of the 12 balls on the left pan, and the same number on the right pan, and push a button to initiate the weighing; there are three possible outcomes: either the weights are equal, or the balls on the left are heavier, or the balls on the left are lighter. Your task is to design a strategy to determine which is the *odd ball* and whether it is *heavier* or *lighter* than the others *in as few uses of the balance as possible*.

While thinking about this problem, you may find it helpful to consider the following questions:

- (a) How can one measure *information*?
- (b) When you have identified the odd ball and whether it is heavy or light, how much information have you gained?
- (c) Once you have designed a strategy, draw a tree showing, for each of the possible outcomes of a weighing, what weighing you perform next. At each node in the tree, how much information have the outcomes so far given you, and how much information remains to be gained?
- (d) How much information is gained when you learn (i) the state of a flipped coin; (ii) the states of two flipped coins; (iii) the outcome when a four-sided die is rolled?
- (e) How much information is gained on the first step of the weighing problem if 6 balls are weighed against the other 6? How much is gained if 4 are weighed against 4 on the first step, leaving out 4 balls?

### Solution:

- (a) Use the entropy definition. Then, there are 24 possible situations (12 balls  $\times$  2 for lighter or heavier) with equal probability so

$$H(X) = \sum \frac{1}{24} \log 24 = \log 24$$

- (b) The information gained is defined as

$$I(X) = H(X) - H(X|E)$$

When odd ball identified,

$$\begin{aligned} I(X) &= H(X) - H(X|E) \\ &= \log 24 - \log 2 \\ &= \log 12 \end{aligned}$$

When whether heavier or lighter identified,

$$\begin{aligned} I(X) &= H(X) - H(X|E) \\ &= \log 24 - \log 12 \\ &= \log 2 = 1 \end{aligned}$$

- (c) Assign each ball with the number 1, 2, ... 12.  
 Step 1: Balance 1-4 against 5-8. If left pan heavier, go 2a. If right pan heavier, go 2b. Else, go 2c.  
 Step 2a: Balance 1,2,5 against 3,4,6. If left pan heavier, balance 1 against 2. If right pan heavier, balance 3 against 4. Else, balance 7 against 8.  
 Step 2b: Balance 1,5,6 against 3,7,8. If left pan heavier, balance 5 against 6. If right pan heavier, balance 7 against 8. Else, balance 3 against 4.  
 Step 2c: balance 1,2,3 against 9, 10, 11. If either side heavier, balance 9 against 10. Else, balance any from 1-11 against 12.

- (d) (i)  $\log 2 - 0 = 1$   
 (ii)  $\log 4 - 0 = 2$   
 (iii)  $\log 4 - 0 = 2$
- (e) If 6 balls are weighed against each other, from either result we have 12 possible situations so the information gained is

$$\log 24 - \log 12 = 1$$

If 4 balls are weighed against each other, from any result we have 8 possible situations so the information gained is

$$\log 24 - \log 8 = \log 3$$

#### Exercise 4.2

Show that, if  $x$  and  $y$  are independent, the entropy of the outcome  $x, y$  satisfies

$$H(X, Y) = H(X) + H(Y). \quad (4.7)$$

In words, entropy is additive for independent variables.

**Solution:**

$$\begin{aligned}
H(X, Y) &= \sum_{i,j} p_i p_j \log \frac{1}{p_i p_j} \\
&= \sum_{i,j} p_i p_j \log \frac{1}{p_i} + \sum_{i,j} p_i p_j \log \frac{1}{p_j} \\
&= \sum_i p_i \log \frac{1}{p_i} + \sum_j p_j \log \frac{1}{p_j} \quad (\because \sum_x p_x = 1) \\
&= H(X) + H(Y)
\end{aligned}$$

#### Exercise 4.4

By how much could the size of a file be reduced given that it is an ASCII file? How would you achieve this reduction?

#### Solution:

We could make it  $\frac{7}{8}$  by eliminating the MSB.

#### Exercise 4.5

Could there be a compressor that maps an outcome  $x$  to a binary code  $c(x)$ , and a decompressor that maps  $c$  back to  $x$ , such that *every possible outcome* is compressed into a binary code of length shorter than  $H_0(X)$  bits?

#### Solution:

No. We need at least  $|A_X|$  distinct states in  $c$  to be decompressed back to  $x$ . If we let  $L = \lceil \log(|A_X|) \rceil - 1 < H_0(X)$ , Then

$$2^L = 2^{\lceil \log(|A_X|) \rceil - 1} < 2^{\log(|A_X|)} = |A_X|$$

which means there are not enough states if the length of compressed binary code is shorter than  $x$ .

#### Exercise 4.8

What are the mathematical shapes of the curves between the cusps?

#### Solution:

As  $\delta$  increases, within the same cusp  $|S_\delta|$  decreases at the same rate.

$$H_\delta(X) = \log |S_\delta|$$

so the graph (within the same cusp) will be the shape of  $y = \log(n - x)$  where

$n$  constant (logarithmic).

#### Exercise 4.9

While some people, when they first encounter the weighing problem with 12 balls and the three-outcome balance (exercise 4.1 (p.66)), think that weighing six balls against six balls is a good first weighing, others say ‘no, weighing six against six conveys *no* information at all’. Explain to the second group why they are both right and wrong. Compute the information gained about *which is the odd ball*, and the information gained about *which is the odd ball and whether it is heavy or light*.

#### Solution:

They are right in the sense that which ball is odd or whether it is heavier or lighter is undetermined at all (all have same probability) so 0 bit gained for each category.

However, there is information gained with respect to which ball is odd **and** heavier or lighter. Half the combinations out of 24 is eliminated, so 1 bit of information gained.

#### Exercise 4.10

Solve the weighing problem for the case where there are 39 balls of which one is known to be odd.

#### Solution:

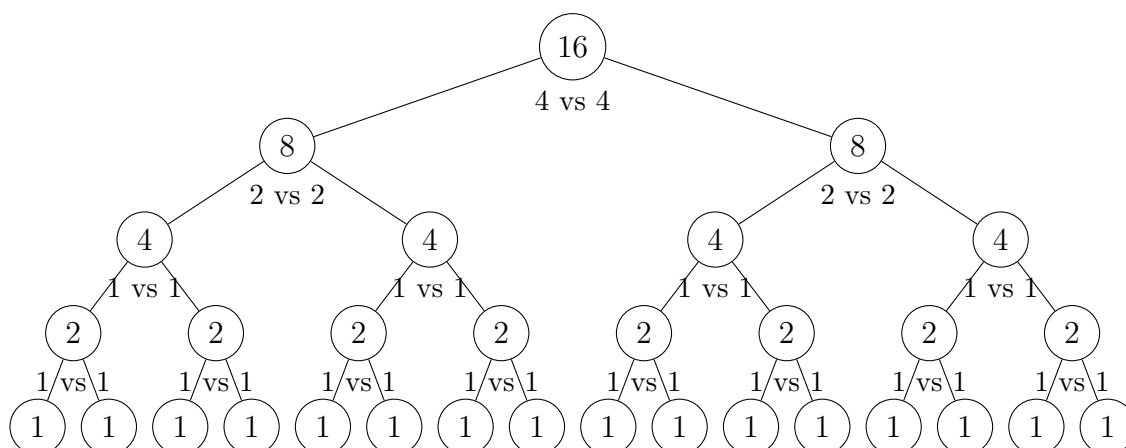
Solve similarly to how we solved question 4.1. For the worst case, weight 13 balls against each other, then 4 balls, then 2 balls, then 1 ball, which takes 4 weighings.

#### Exercise 4.11

You are given 16 balls, all of which are equal in weight except for one that is either heavier or lighter. You are also given a bizarre two-pan balance that can report only two outcomes: ‘the two sides balance’ or ‘the two sides do not balance’. Design a strategy to determine which is the odd ball in as few uses of the balance as possible.

#### Solution:

There are only two outcomes per weighing, so we can only gain at most 2 bits per weighing.



As above, 4 weighings are needed.

#### Exercise 4.12

You have a two-pan balance; your job is to weigh out bags of flour with integer weights 1 to 40 pounds inclusive. How many weights do you need? [You are allowed to put weights on either pan. You're only allowed to put one flour bag on the balance at a time.]

#### Solution:

We can solve this problem using binary search.

For example, if in all weighings the bags of flour was heavier, we can weigh against 20, 30, 35, 38, 39 in that order.

We need 5 weights.

#### Exercise 4.13

- Is it possible to solve exercise 4.1 (p.66) (the weighing problem with 12 balls and the three-outcome balance) using a sequence of three *fixed* weighings, such that the balls chosen for the second weighing do not depend on the outcome of the first, and the third weighing does not depend on the first or second?
- Find a solution to the general  $N$ -ball weighing problem in which exactly one of  $N$  balls is odd. Show that in  $W$  weighings, an odd ball can be identified from among  $N = (3^W - 3)/2$  balls.

#### Solution:

- Yes. We can label each ball as AAB, ABA, ABB, ABC, BBC, BCA, BCB, BCC, CAA, CAB, CAC, CCA. Then, for  $i$ th weighing, for each

ball look at  $i - 1$ th letter and if it is A weigh on left pan, if B weigh on right pan, and if C do not use for weighing. Then, take the result of the weighing and if left pan heavier say A, right pan heavier say B, and balanced C (For example, if left pan heavy, balanced, left pan heavy then the sequence for the weighing is BCB). The sequence obtained will uniquely indicate which ball was odd and whether it was heavier or lighter. If the sequence exactly matches, the ball is odd and heavier and if the sequence matches in a inverted way (e.g. BBA match AAB invertedly), the ball is odd and lighter. The sequence for the ball is chosen in a way such that at each weighing 4 balls are weighed against 4 balls, all balls are weighed (no CCC), no balls that are always weighed against each other which gives no distinction of heavy/light between them (AAA, BBB) and none are inverse of another (the first letter change is always  $A \rightarrow B$ ,  $B \rightarrow C$  or  $C \rightarrow A$ ).

- (b) We can have  $3^W$  results from the weighing, and this can be corresponded to each ball as in (a). As described in (a), we must not use AAA, BBB and CCC, requiring subtraction by 3. Also, we must remove inverses since ABC heavy and CBA light will give the same result, which requires dividing by 2. Hence we get  $N = \frac{3^W - 3}{2}$ .

#### Exercise 4.14

You are given 12 balls and the three-outcome balance of exercise 4.1; this time, *two* of the balls are odd; each odd ball may be heavy or light, and we don't know which. We want to identify the odd balls and in which direction they are odd.

- (a) Estimate how many weighings are required by the optimal strategy. And what if there are three odd balls?
- (b) How do your answers change if it is known that all the regular balls weigh 100 g, that light balls weigh 99 g, and heavy ones weigh 110 g?

#### Solution:

- (a) For 2 balls,  $\binom{12}{2} \times 4 = 264$ ,  $3^5 < 264 < 3^6$  so 6 times.

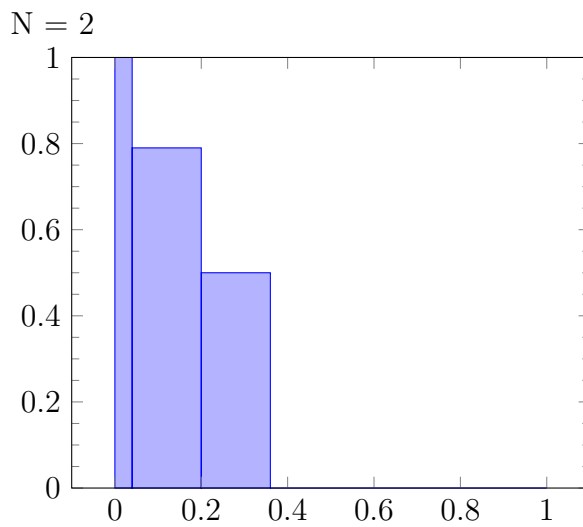
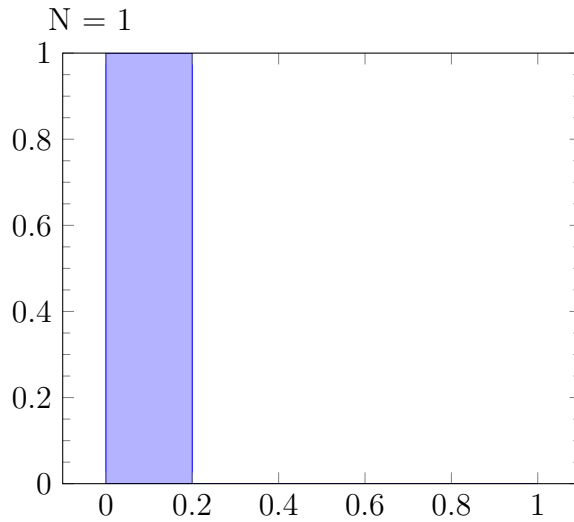
For 3 balls,  $\binom{12}{3} \times 2^3 = 1760$ ,  $3^6 < 1760 < 3^7$  so 7 times.

- (b) The theoretical lower bound will still be the same, but practically the solution will be easier to find since "cancelling out" will not happen.

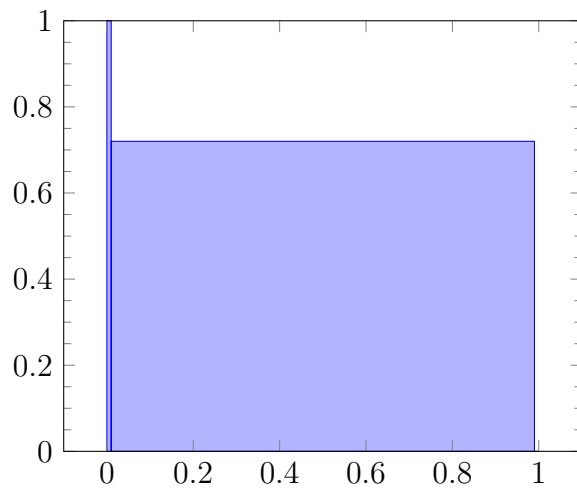
### Exercise 4.15

Let  $\mathcal{P}_X = \{0.2, 0.8\}$ . Sketch  $\frac{1}{N}H_\delta(X^N)$  as a function of  $\delta$  for  $N = 1, 2$  and 1000.

**Solution:**



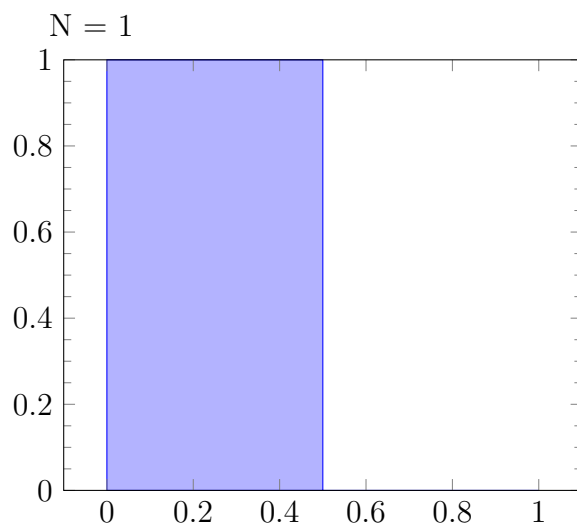
N = 1000 (very rough approximate, flatten at around 0.722)



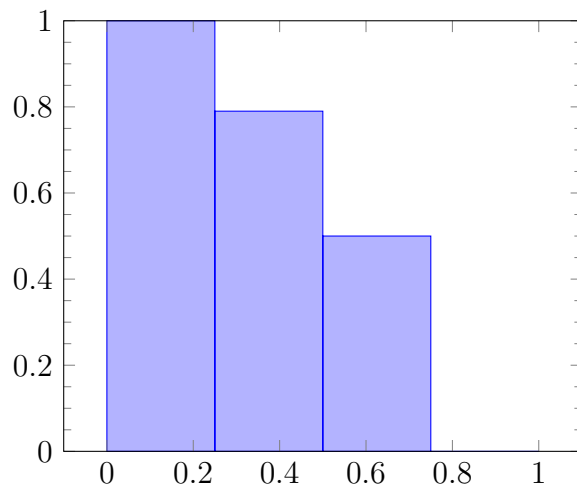
#### Exercise 4.16

Let  $\mathcal{P}_Y = \{0.5, 0.5\}$ . Sketch  $\frac{1}{N}H_\delta(Y^N)$  as a function of  $\delta$  for  $N = 1, 2, 3$  and 100.

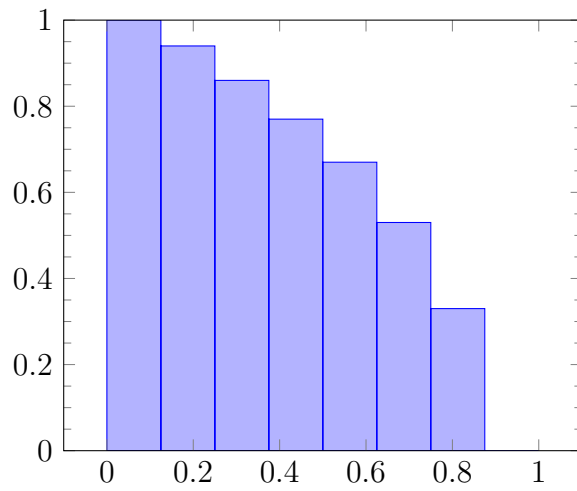
**Solution:**



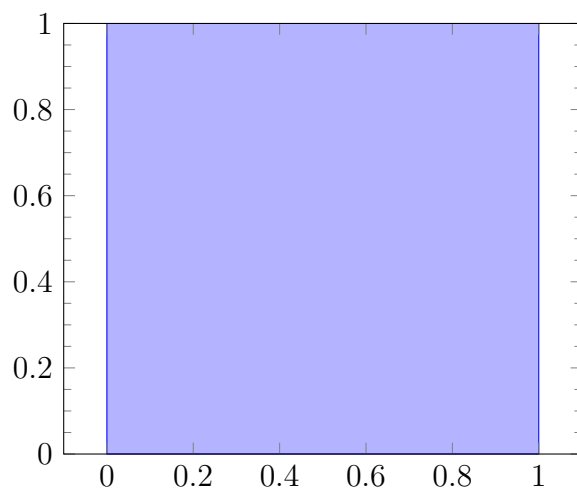
N = 2



$N = 3$



$N = 100$  (rough approximation)



### Exercise 4.17

(For physics students.) Discuss the relationship between the proof of the ‘asymptotic equipartition’ principle and the equivalence (for large systems) of the Boltzmann entropy and the Gibbs entropy.

#### Solution:

Gibbs entropy:  $S_G = k_B \sum_i p_i \ln \frac{1}{p_i}$ , which is equivalent to the Shannon entropy apart from the constant factor.

Boltzmann entropy is  $S_B = k_B \ln \Omega$  where  $\Omega$  is the number of accessible states in a microcanonical ensemble. This is equivalent to the perfect information content  $H_0$  of the ensemble apart from the constant factor.

Now, consider a thermal distribution

$$P(x) = \frac{1}{Z} \exp\left(-\frac{E(x)}{k_B T}\right)$$

Then,

$$\ln P(x) = -\ln Z - \frac{E(x)}{k_B T}$$

This means fixing energy is equivalent to fixing the probability. Now, the AEP states that for a large number of independent components, the outcome of a certain process will almost certainly fall in the typical set.

If we set the energy of the microcanonical ensemble to the mean energy of the canonical ensemble, the microcanonical ensemble and the uniform distribution over the typical set of the canonical ensemble becomes identical due to AEP. Hence, for large systems  $S_G \approx S_B$ , and for thermodynamic limit ( $N \rightarrow \infty$ ),  $S_G = S_B$ .

### Exercise 4.18

Sketch the Cauchy distribution

$$P(x) = \frac{1}{Z} \frac{1}{x^2 + 1}, \quad x \in (-\infty, \infty). \quad (4.40)$$

What is its normalizing constant  $Z$ ? Can you evaluate its mean or variance?

Consider the sum  $z = x_1 + x_2$ , where  $x_1$  and  $x_2$  are independent random variables from a Cauchy distribution. What is  $P(z)$ ? What is the probability distribution of the mean of  $x_1$  and  $x_2$ ,  $\bar{x} = (x_1 + x_2)/2$ ? What is the probability distribution of the mean of  $N$  samples from this Cauchy distribution?

#### Solution:

Sketch can be found here: <https://www.desmos.com/calculator/bchpkfotxz>

Finding the normalizing constant:

$$\begin{aligned}\int_{-\infty}^{\infty} \frac{1}{Z} \frac{1}{x^2 + 1} dx &= 1 \\ \int_{-\infty}^{\infty} \frac{1}{x^2 + 1} dx &= Z \\ [\arctan(x)]_{-\infty}^{\infty} &= Z \\ \frac{\pi}{2} - (-\frac{\pi}{2}) &= Z \\ Z &= \pi\end{aligned}$$

$$\begin{aligned}E[X] &= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{x}{x^2 + 1} dx \\ &= \frac{1}{\pi} \left\{ \int_0^{\infty} \frac{x}{x^2 + 1} dx + \int_{-\infty}^0 \frac{x}{x^2 + 1} dx \right\} \\ &= \frac{1}{\pi} \{ \infty - \infty \}\end{aligned}$$

so the mean is undefined, which also implies the variance is undefined.

The PDF of Cauchy(scale= $\sigma$ ) is

$$P(x) = \frac{1}{\pi} \frac{\sigma}{\sigma^2 + x^2}$$

and the characteristic function is

$$\phi(t) = \exp(-\sigma|t|)$$

Then, the characteristic function of  $z = x_1 + x_2$  is

$$\phi_z(t) = \phi_{x_1}(t) \cdot \phi_{x_2}(t) = \exp(-\sigma|t|) \cdot \exp(-\sigma|t|) = \exp(-2\sigma|t|)$$

This is the characteristic function of Cauchy(scale= $2\sigma$ ), so  $z$  is Cauchy distributed with scale  $2\sigma$ .

Similarly, the characteristic function of  $\bar{x} = (x_1 + x_2)/2$  is

$$\phi_{\bar{x}}(t) = \phi_z(t/2) = \exp(-\sigma|t|)$$

This is the characteristic function of Cauchy(scale= $\sigma$ ), so  $\bar{x}$  is also Cauchy distributed with scale  $\sigma$ .

Similarly, the characteristic function of  $\bar{x}_N = (x_1 + x_2 + \dots + x_N)/N$  is

$$\phi_{\bar{x}_N}(t) = (\phi_x(t/N))^N = (\exp(-\sigma|t/N|))^N = \exp(-\sigma|t|)$$

This is the characteristic function of Cauchy(scale= $\sigma$ ), so  $\bar{x}_N$  is also Cauchy distributed with scale  $\sigma$ .

Hence, the mean of  $N$  samples from a Cauchy distribution is also Cauchy distributed with the same scale parameter  $\sigma$  as the original distribution, and the mean does not converge to a single value as  $N$  increases, unlike distributions with finite variance.

### Exercise 4.19

Chernoff bound. We derived the weak law of large numbers from Chebyshev's inequality (4.30) by letting the random variable  $t$  in the inequality  $P(t \geq \alpha) \leq \bar{t}/\alpha$  be a function,  $t = (x - \bar{x})^2$ , of the random variable  $x$  we were interested in.

Other useful inequalities can be obtained by using other functions. The Chernoff bound, which is useful for bounding the tails of a distribution, is obtained by letting  $t = \exp(sx)$ .

Show that

$$P(x \geq a) \leq e^{-sa}g(s), \quad \text{for any } s > 0 \quad (4.41)$$

and

$$P(x \leq a) \leq e^{-sa}g(s), \quad \text{for any } s < 0 \quad (4.42)$$

where  $g(s)$  is the moment-generating function of  $x$ ,

$$g(s) = \sum_x P(x)e^{sx}. \quad (4.43)$$

**Solution:**

### Exercise 4.20

This exercise has no purpose at all; it's included for the enjoyment of those who like mathematical curiosities.

Sketch the function

$$f(x) = x^{x^{x^{x^{\dots}}}} \quad (4.44)$$

for  $x \geq 0$ . Hint: Work out the inverse function to  $f$  – that is, the function  $g(y)$  such that if  $x = g(y)$  then  $y = f(x)$  – it's closely related to  $p \log 1/p$ .

**Solution:**

Let  $y = f(x)$ . Then, the equation is  $y = x^y$ . Rearranging, we get  $\log(x) = \frac{1}{y} \log(y)$ .